

$$\frac{d}{dx} \sin(x) = \cos(x) \quad \frac{d}{dx} \cos(x) = -\sin(x) \quad \frac{d}{dx} \tan(x) = \sec^2(x) \quad \frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x) \quad \frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\int \cos(x) dx = \sin(x) + C \quad \int \sin(x) dx = -\cos(x) + C \quad \int \sec^2(x) dx = \tan(x) + C$$

$$\int \sec(x) \tan(x) dx = \sec(x) + C \quad \int \csc(x) \cot(x) dx = -\csc(x) + C \quad \int \csc^2(x) dx = -\cot(x) + C$$

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx} \cos^{-1}(x) = \frac{(-1)}{\sqrt{1-x^2}} \quad \frac{d}{dx} \tan^{-1}(x) = \frac{1}{(1+x^2)} \quad \frac{d}{dx} \cot^{-1}(x) = \frac{(-1)}{(1+x^2)}$$

$$\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}} \quad \frac{d}{dx} \csc^{-1}(x) = \frac{(-1)}{x\sqrt{x^2-1}}$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C \quad \int \frac{1}{\sqrt{1+x^2}} dx = \tan^{-1} x + C \quad \int \frac{1}{x} \sqrt{x^2-1} dx = \sec^{-1} x + C$$

$$b^{-(n)} = \frac{1}{b^n} \quad \ln(x^n) = n \ln(x) \quad \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c \quad \int u dv = uv - \int v du \quad \ln^n(x) = (\ln x)^n$$

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st} dt$$

$$\mathcal{L}\{c\} = \frac{\{c\}}{s} \quad \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad \mathcal{L}\{e^{\{c\}t}\} = \frac{1}{s-\{c\}} \quad \mathcal{L}\{\sin(kt)\} = \frac{k}{s^2+k^2} \quad \mathcal{L}\{\cos(kt)\} = \frac{s}{s^2+k^2}$$

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0) \quad \mathcal{L}\{f''(t)\} = s^3F(s) - s^2f(0) - sf'(0) - f''(0)$$

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x) \quad \sin^2(x) = \frac{1-\cos^2(x)}{2} \quad \cos^2(x) = \frac{1+\cos^2(x)}{2}$$

$$\mathcal{L}\{e^{at} \cdot f(t)\} = F(s-a)$$